

IAN COLLIARD, PH.D.
LIVERMORE, CA, USA | COLLIARD1@LLNL.GOV | +1 (503) 863-0362
[LINKEDIN](#) | [GOOGLE SCHOLAR](#) | [PERSONAL WEBSITE](#)
LANGUAGES: ENGLISH (NATIVE), SPANISH (NATIVE), FRENCH (PROFICIENT)

LABORATORY & TECHNICAL SKILLS

Synthesis & Handling: Air-sensitive techniques; aqueous & microscale; actinide/lanthanide complexes; metallic aerogels; Polyoxometalate synthesis & functionalization; nanowire synthesis; radioactive material worker (Th, U, Np, Pu, Am243, Cm248, Cf249, Ra226)

Characterization: SCXRD (>219 submitted), PXRD, PDF, SAXS; IR, Raman, luminescence spectroscopy; multinuclear NMR (^1H , ^{13}C , ^{31}P , ^{119}Sn , ^{183}W); SEM, AFM, X-ray tomography, Cyclic Voltammetry, etc.

Instrumentation & Management: XRD facility management (Rigaku, Bruker); maintenance, service requests, financial oversight; user training, safety compliance, inventory, waste management, scheduling coordinator,

Software & Data Analysis: Crystallographic software (OLEX2, Chrysalis, JANA); data analysis & reporting (Igor, Origin, Excel, Word); Coding (Python); Graphic design (CorelDraw, inkscape).

EDUCATION

Ph.D., Materials Chemistry, Oregon State University, Corvallis, OR (2017 – 2022)

Advisor: Dr. May Nyman; *Dissertation focus:* Metal-oxo cluster chemistry of actinides (Th, U) and Ce; synthesis and structural characterization of polyoxometalate complexes.

B.S., Chemistry Fordham University, Bronx, NY (2014 – 2017)

RESEARCH EXPERIENCE

Lawrence Livermore National Laboratory (LLNL) – Postdoctoral Researcher & XRD Facility Manager (2022–Present)

Advisors: Dr. Gauthier Deblonde & Dr. Tyler Fears

- Pioneered micro-scale synthesis of lanthanide and transplutonium polyoxometalate complexes, enabling first structural characterization of curium–POM compounds at the microgram scale (*Nature Chemistry*, *ChemComm*).
- Developed synthetic strategies to modulate optical and spectroscopic properties of *f*-element complexes, revealing coordination trends between lanthanides and actinides (*JACS Au*).
- Fabricated multi-component metallic aerogels (Ag, Au) via novel routes, enhancing structural stability and catalytic activity.
- Managed and optimized three X-ray diffraction instruments (Rigaku SCXRD, Bruker D8, D2 Phaser) supporting 20+ research groups annually; led safety oversight, user training, and scheduling.
- Solved and refined 100+ crystal structures in support of multi-lab, multi-disciplinary projects.

Lawrence Livermore National Laboratory – DOE SCGSR and GEM Fellow (Feb – Aug 2021)

Advisor: Dr. Gauthier Deblonde

- Designed and implemented synthetic and spectroscopic methods for transplutonium–POM complexes, culminating in the isolation of novel curium coordination compounds (*Nature Chemistry*)

Oregon State University, Corvallis, OR – Graduate Research Assistant (Jan 2018 – March 2022)

Advisor: Dr. May Nyman

- Expanded aqueous actinide chemistry (Th, U, Ce) through controlled cation-mediated synthesis, isolating clusters from 1–70 metal centers and advancing understanding of metal–oxo assembly.
- Synthesized and functionalized tungsten polyoxometalates with organotin ligands to enhance actinide binding and tune solubility.
- Applied SCXRD, SAXS, and multinuclear NMR to establish structure–property relationships, contributing to 10+ peer-reviewed publications.

Fordham University, Bronx, NY – Undergraduate Research Assistant (Jan 2016 – June 2017)

Advisor: Prof. Christopher Koenigsmann

- Investigated bi-metallic nanowire synthesis for electrochemical glucose sensing, optimizing core–shell fabrication and electrochemical performance.

TOP 3 PUBLICATION

I. Colliard and G. Deblonde, Similar but different: structural and spectroscopic characterization of a series of europium and curium coordination complexes, *Journal of the American Chemical Society* **2025**, 147 (17), 14455–14467. <https://doi.org/10.1021/jacs.5c00861>

I. Colliard and G. Deblonde, *Polyoxometalate ligands reveal different coordination chemistries among lanthanides and heavy actinides*, *JACS Au* **2024**, 4 (7), 2503–2513. <https://doi.org/10.1021/jacsau.4c00245>

I. Colliard, et al, Polyoxometalates as ligands to synthesis, isolate and characterize compounds of rare isotopes on the microgram scale, *Nature Chemistry* **2022**, 14 (12), 1357–1366. <https://doi.org/10.1021/acs.inorgchem.2c04014>

FELLOWSHIPS AND AWARDS

- **PLS Directorate Outstanding Postdoc Award**, LLNL – 2025
- **Lindau Nobel Laureate Fellowship**, LLNL – 2025
- **PLS Directorate Winter Scientific & Technical Award**, LLNL – 2024
- **Director’s Excellence in Publication Award**, LLNL – 2023
- **PLS Directorate Winter Publication Award**, LLNL – 2023
- **GEM Consortium Fellowship**, National GEM Consortium – 2019
- **DOE SCGSR Fellowship**, U.S. Department of Energy – 2019
- **NSF GRFP Honorable Mention**, National Science Foundation – 2018

CONFERENCE PRESENTATIONS

Oral Presentations

- *Polyoxometalate ligands reveal different coordination chemistries among lanthanides and heavy actinides* – ACS National Meeting, San Francisco, CA (2023)
- *Actinide–Oxide Cluster: Structure and Properties* – Actinide Center of Excellence Meeting, Pojoaque, NM (2018)
- *Core-Shell Platinum Coated Nickel Nanowires as Electrochemical Catalysts for Glucose Sensors* – Undergraduate Research Symposium, Fordham University, NY (2017)

Poster Presentations

- *Effects of Metal Ions on Uranium(IV) Sulfate Speciation and Supra-molecular Assembly* – FMOCS VI, Corvallis, OR (2019)
- *Synthesis and Characterization of Novel Uranium Sulfate Frameworks: From Simple Dimeric Species to Hetero-Metallic Frameworks* – ACS NORM, Portland, OR (2019)
- *Synthesis and Investigation of Tetravalent Uranium Sulfate Speciation* – NNSA SSAP Symposium, Albuquerque, NM (2019)

- *Synthesis and Investigation of Mono- and Poly-nuclear Actinide POM Complexes* – SACNAS National Conference, San Antonio, TX (2018)
- *Synthesis and Investigation of Mononuclear Actinide POM Complexes* – ACS Northwest Regional Meeting, Richland, WA (2018)

TEACHING AND OUTREACH EXPERIENCE

- Graduate School Ambassador, Oregon State University – 2018–2022
- SACNAS Volunteer Mentor, Oregon State University – 2018–2022
- Chair and Organizer, SACNAS Symposium "Opportunities in Actinide Chemistry" – 2019
- Teaching Assistant, Oregon State University – 2017
- Volunteer Tutor, Fordham University – 2016–2017

PROFESSIONAL ASSOCIATIONS

- American Chemical Society
- IUCr Crystallographer
- Society of the Advancement of Chicano and Native American in the Sciences
- Sigma Xi

REFERENCES

- Dr. Gauthier Deblonde — Lawrence Livermore National Laboratory, deblonde1@llnl.gov
- Prof. Dr. May Nyman — Oregon State University, may.nyman@oregonstate.edu

PUBLICATIONS LIST

- [28] J. Bustos, **I. Colliard**, et al, Structural and Spectroscopic Characterization of Plutonium and Other Tetravalent Metals Complexed to a Keggin Ion, **2025**, XXX-XXX. <https://doi.org/10.1021/acs.inorgchem.5c03627>
- [27] A. Hastings, **I. Colliard**, et al, Breaking the Neptunyl Barrier: Direct Access to Neptunium(IV) in Aqueous Solution via Polyoxometalate-Mediated Reduction and Stabilization. *Inorganic Chemistry*, **2025**, 64 (37), 18925–18937. <https://doi.org/10.1021/acs.inorgchem.5c02949>
- [26] **I. Colliard**, V. Stavila, G. Deblonde, Isolation of a New Polyoxometalate Complex of Plutonium. *Chemical Communications* **2025**, 61 (63), 11858–11861. <https://doi.org/10.1039/D5CC03481G>
- [25] M. Shohel, **I. Colliard**, M. Nyman, Pertechetate/Perrhenate-Linked Frameworks of UIV , UIV 6-Hexamer and Uranyl (VI) Building Units . *Inorganic Chemistry* **2025**, 64 (29), 15039–15049. <https://doi.org/10.1021/acs.inorgchem.5c01958>
- [24] **I. Colliard** and G. Deblonde, Similar but different: structural and spectroscopic characterization of a series of europium and curium coordination complexes, *Journal of the American Chemical Society* **2025**, 147 (17), 14455–14467. <https://doi.org/10.1021/jacs.5c00861>
- [23] **I. Colliard** and G. Deblonde, Isolation and characterization of a new polyoxometalate ligand, H₃SbW₁₄O₅₀¹⁰⁻, and its interactions with *f*-elements, *Chemical Communications* **2025**, 61

- (5), 941–944. <https://doi.org/10.1039/D4CC05558F>
- [22] **I. Colliard** and G. Deblonde, *Crystal structure of hexachlorothallate within a caesium chloride phosphotungstate lattice* $\text{Cs}_9(\text{TlCl}_6)(\text{PW}_{12}\text{O}_{40})_2 \cdot 9\text{CsCl}$, *Acta Crystallogr E Crystallogr Commun* **2024**, 80 (Pt 7), 721–724. <https://doi.org/10.1107/S2056989024005565>
- [21] **I. Colliard** and G. Deblonde, From +I to +IV, alkalis to actinides: Capturing cations across the periodic table with Keggin polyoxometalate ligands. *Inorganic Chemistry*, **2024**, 63 (35), 16293–16303. <https://doi.org/10.1021/acs.inorgchem.4c02254>
- [20] **I. Colliard** and G. Deblonde, *Synthesis and structural characterization of a hydrated sodium–caesium tetracosatungstate(VI)*, $\text{Na}_5\text{Cs}_{19}[\text{W}_{24}\text{O}_{84}] \cdot 21\text{H}_2\text{O}$, *Acta Crystallogr E Crystallogr Commun* **2024**, 80 (Pt 6), 667–670. <https://doi.org/10.1107/S2056989024004778>
- [19] **I. Colliard** and G. Deblonde, *Isolation and characterization of a Peacock-Weakley polyoxometalate containing a transplutonium element: the case of curium bis-pentatungstate* $[\text{Cm}(\text{W}_5\text{O}_{18})_2]^{9-}$, *Chemical Communications* **2024**, 60 (47), 5999–6002. <https://doi.org/10.1039/D4CC01381F>
- [18] **I. Colliard** and G. Deblonde, *Polyoxometalate ligands reveal different coordination chemistries among lanthanides and heavy actinides*, *JACS Au* **2024**, 4 (7), 2503–2513. <https://doi.org/10.1021/jacsau.4c00245>
- [17] M. Nyman, T. Rahman, **I. Colliard**, *Decaniobate: the fruitfly of niobium polyoxometalate chemistry*, *Acc. Chem. Res.* **2023**, 56 (24), 3616–3625. <https://doi.org/10.1021/acs.accounts.3c00583>
- [16] C. Colla, **I. Colliard**, et al, *Contrasting Trivalent Lanthanide and Actinide Complexation by Polyoxometalates via solution-state NMR*, *Inorganic Chemistry*, **2023**, 62 (16), 6242–6254. <https://doi.org/10.1021/acs.inorgchem.2c04014>
- [15] **I. Colliard**, et al, *Polyoxometalates as ligands to synthesis, isolate and characterize compounds of rare isotopes on the microgram scale*, *Nature Chemistry* **2022**, 14 (12), 1357–1366. <https://doi.org/10.1021/acs.inorgchem.2c04014>
- [14] M. C. Wasson, X. Wang, P. Melix, S. Alayoglu, A. T. Y. Wolek, **I. Colliard**, et al, *Interfacial Unit-Dependent Catalytic Activity for CO Oxidation over Cerium Oxysulfate Cluster Assemblies*, *ACS Appl Mater Interfaces* **2022**, 14 (29), 33515–33524. <https://doi.org/10.1021/acsami.2c05937>
- [13] **I. Colliard**, et al, *Metal–Oxo Cluster Formation Using Ammonium and Sulfate to Differentiate M^{IV} (Th, U, Ce) Chemistries*, *Inorganic Chemistry*, **2023**, 62 (5), 1891–1900. <https://doi.org/10.1021/acs.inorgchem.2c01309>
- [12] J. Chen, L. Vilà-Nadal, A. Solé-Daura, G. Chisholm, T. Minato, C. Busche, T. Zhao, B. Kandasamy, A. Y. Ganin, R. M. Smith, **I. Colliard**, et al, *Effective Storage of Electrons in Water by the Formation of Highly Reduced Polyoxometalate Clusters*, *Journal of the American Chemical Society*, **2022**, 144 (20), 8951–8960.

<https://doi.org/10.1021/jacs.1c10584>

- [11] R. Smith, **I. Colliard**, etc, Silver–Cytosine–Polyoxometalate Assemblies: Assessing the Role of Polyoxometalates in Constructing Ag–DNA Suprastructures, *Crystal Growth & Design*, **2022**, 22 (4), 2294–2306. <https://doi.org/10.1021/acs.cgd.1c01385>
- [10] X. Wang, K. Brunson, H. Xie, **I. Colliard**, et al, Heterometallic Ce^{IV}/V^V Oxo Clusters with Adjustable Catalytic Reactivities, *Journal of the American Chemical Society*, **2021**, 143, 49, 21056–21065. <https://doi.org/10.1021/jacs.1c11208>
- [9] **I. Colliard**, et al, Snapshots of Ce₇₀ Toroid Assembly from Solids and Solution, *Journal of the American Chemical Society*, **2021**, 143 (25), 9612–9621. <https://doi.org/10.1021/jacs.1c04095>
- [8] M. Olsen, **I. Colliard**, et al, Hybrid Polyoxometalate Salt Adhesion by Butyltin Functionalization, *ACS Appl Mater Interfaces* **2021**, 13 (16), 19497–19506. <https://doi.org/10.1021/acsami.1c03269>
- [7] **I. Colliard**, M. Nyman, Ce^{IV}₇₀ Oxosulfate Rings, Frameworks, Supramolecular Assembly, and Redox Activity, *Angewandte Chemie International Edition* **2021**, 60 (13), 7308–7315. <https://doi.org/10.1002/anie.202016522>
- [6] **I. Colliard**, et al, Bridging the Transuranics with Uranium(IV) Sulfate Aqueous Species and Solid Phases, *Inorganic Chemistry*, **2020**, 59 (23), 17049–17057. <https://doi.org/10.1021/acs.inorgchem.0c02267>
- [5] **I. Colliard**, M. Nyman, Building [U^{IV}₇₀(OH)₃₆(O)₆₄]^{4−} Oxocluster Frameworks with Sulfate, Transition Metals, and U^V, *Chemistry – A European Journal* **2020**, 26 (54), 12481–12488. <https://doi.org/10.1002/chem.202002403>
- [4] **I. Colliard**, et al, Supramolecular Assembly of U(IV) Clusters and Superatoms with Unconventional Counteranions, *Journal of the American Chemical Society*, **2020**, 142 (19), 9039–9047. <https://doi.org/10.1021/jacs.0c03041>
- [3] J.N. Wacker, M. Vasiliu, **I. Colliard**, et al, Monomeric and Trimeric Thorium Chlorides Isolated from Acidic Aqueous Solution, *Inorganic Chemistry*, **2019**, 58 (16), 10871–10882. <https://doi.org/10.1021/acs.inorgchem.9b01238>
- [2] A Vanagas, J.N. Wacker, C.L. Rom; E.N. Glass, **I. Colliard**, et al, Solution and Solid State Structural Chemistry of Th(IV) and U(IV) 4-Hydroxybenzoates, *Inorganic Chemistry* **2018**, 57 (12), 7259–7269. <https://doi.org/10.1021/acs.inorgchem.8b00919>
- [1] C. Koenigsmann, and **I. Colliard**, *One-Dimensional Nanostructures for PEM Fuel Cell Applications*, in Academic Press, **2017**. 19–48.